http://localhost:8000

http://localhost:8000/docs

http://localhost:8000/openapi.json

/

/health

/employees

/employees/{employee_id}

/products

/products/{product_code}

/saved-queries

/saved-queries

/saved-queries/{query_id}

/saved-queries/{query_id}

/saved-queries/{query_id}

/analytics/store-sales-summary

/analytics/product-ranking

/analytics/employee-performance

/analytics/sales-trend

/analytics/analyze

/analytics/logs

/analytics/logs/{log_id}/turns

/skills/observe_schema

/skills/run_python_analysis

X-API-Key .env → API_KEY

insightagent-secret-key

```
curl http://localhost:8000/employees \
  -H "X-API-Key: insightagent-secret-key"
```

2xx

401 {"error": "invalid_api_key", "message": "Invalid or missing X-API-Key header."}

API_KEY

GET / GET /health

200	
201	/saved-queries
204	
400	
401	X-API-Key
404	
422	
500	

```
{
  "detail": {
    "error": "invalid_api_key",
    "message": "Invalid or missing X-API-Key header."
  }
}
```

GET /employees/not-a-number

```
{
  "detail": [
    {
      "type": "int_parsing",
      "loc": ["path", "employee_id"],
      "msg": "Input should be a valid integer, unable to parse string
        as an integer",
      "input": "not-a-number"
    }
```

```
  ]
}
```


GET /products?limit=0


```
{
  "detail": [
    {
      "type": "greater_than_equal",
      "loc": ["query", "limit"],
      "msg": "Input should be greater than or equal to 1",
      "input": "0"
    }
  ]
}
```


POST /saved-queries title


```
{
  "detail": [
    {
      "type": "missing",
      "loc": ["body", "title"],
      "msg": "Field required",
      "input": {}
    }
  ]
}
```


/skills/run_python_analysis 200 {"detail": {...}}

	error_type	status
200		success
403	SECURITY_VIOLATION	blocked
403	SQL_BLOCKED	failed
408	EXECUTION_TIMEOUT	failed
422	SYNTAX_ERROR	blocked

	error_type	status
422	RUNTIME_ERROR	failed
500	WORKER_ERROR	failed

GET /

200

```
{
  "status": "ok",
  "service": "InsightAgent API",
  "version": "1.0.0",
  "docs": "/docs",
  "database": "enterprise_api"
}
```

GET /health

200

```
{
  "status": "ok",
  "db_connections": {
    "rw": "ok",
    "ro": "ok"
  }
}
```

200

```
{
  "status": "degraded",
  "db_connections": {
    "rw": "ok",
    "ro": "error: (OperationalError) ..."
  }
}
```

200

employees X-API-Key

GET /employees

skip	0	≥ 0
limit	100	1 – 500

```
curl "http://localhost:8000/employees?skip=0&limit=10" \
  -H "X-API-Key: insightagent-secret-key"
```

200

```
{
  "total": 23,
  "items": [
    {
      "employeeNumber": 1002,
      "lastName": "Murphy",
      "firstName": "Diane",
      "extension": "x5800",
      "email": "dmurphy@classicmodelcars.com",
      "officeCode": "1",
      "reportsTo": null,
      "jobTitle": "President"
    }
  ]
}
```

401 422 500

GET /employees/{employee_id}

employee_id	employeeNumber

```
curl http://localhost:8000/employees/1002 \
  -H "X-API-Key: insightagent-secret-key"
```

200

```
{
  "employeeNumber": 1002,
  "lastName": "Murphy",
  "firstName": "Diane",
  "extension": "x5800",
  "email": "dmurphy@classicmodelcars.com",
  "officeCode": "1",
  "reportsTo": null,
  "jobTitle": "President"
}
```

404

```
{
  "detail": "Employee 9999 not found."
}
```

401 404 422
employee_id 500

products X-API-Key

GET /products

skip	0	≥ 0
limit	100	1 – 500

```
curl "http://localhost:8000/products?skip=0&limit=5" \
  -H "X-API-Key: insightagent-secret-key"
```

200

```
{
  "total": 110,
  "items": [
    {
      "productCode": "S10_1678",
      "productName": "1969 Harley Davidson Ultimate Chopper",
      "productLine": "Motorcycles",
      "productScale": "1:10",
      "productVendor": "Min Lin Diecast",
      "productDescription": "This replica features...",
      "quantityInStock": 7933,
      "buyPrice": 48.81,
      "MSRP": 95.70
    }
  ]
}
```

401 422 500

GET /products/{product_code}

product_code	productCode	S10_1678

```
curl http://localhost:8000/products/S10_1678 \
  -H "X-API-Key: insightagent-secret-key"
```

200

```
{
  "productCode": "S10_1678",
  "productName": "1969 Harley Davidson Ultimate Chopper",
  "productLine": "Motorcycles",
  "productScale": "1:10",
  "productVendor": "Min Lin Diecast",
  "productDescription": "This replica features...",
  "quantityInStock": 7933,
  "buyPrice": 48.81,
  "MSRP": 95.70
}
```

404

```
{
  "detail": "Product INVALID_CODE not found."
}
```

401 404 500

---

saved_queries X-API-Key

POST /saved-queries

---

title	✅
natural_language_query	✅
generated_code	❌
result_summary	❌

```
curl -X POST http://localhost:8000/saved-queries \
  -H "X-API-Key: insightagent-secret-key" \
```

```
  -H "Content-Type: application/json" \
  -d '{
    "title": "Revenue by Product Line",
    "natural_language_query": "What is the total revenue for each
        product line?",
    "generated_code": "df = read_sql(\"SELECT productLine,
        SUM(quantityOrdered * priceEach) AS revenue FROM orderdetails
        od JOIN products p ON od.productCode = p.productCode GROUP BY
        productLine\")\nprint(df)",
    "result_summary": "Classic Cars had the highest revenue at
        $3.85M."
  }'
```

201

```
{
  "id": 1,
  "title": "Revenue by Product Line",
  "natural_language_query": "What is the total revenue for each
        product line?",
  "generated_code": "df = read_sql(...)\nprint(df)",
  "result_summary": "Classic Cars had the highest revenue at $3.85M.",
  "created_at": "2025-04-13T10:00:00",
  "updated_at": "2025-04-13T10:00:00"
}
```

401 422
500

GET /saved-queries

---

skip	0	≥ 0
limit	100	1 – 500

```
curl "http://localhost:8000/saved-queries?skip=0&limit=10" \
  -H "X-API-Key: insightagent-secret-key"
```

200

```
{
  "total": 5,
```

```
  "items": [
    {
      "id": 1,
      "title": "Revenue by Product Line",
      "natural_language_query": "What is the total revenue for each
        product line?",
      "generated_code": "...",
      "result_summary": "Classic Cars had the highest revenue.",
      "created_at": "2025-04-13T10:00:00",
      "updated_at": "2025-04-13T10:00:00"
    }
  ]
}
```

401 422 500

GET /saved-queries/{query_id}

---

query_id

```
curl http://localhost:8000/saved-queries/1 \
  -H "X-API-Key: insightagent-secret-key"
```

200

404

```
{
  "detail": "SavedQuery 999 not found."
}
```

401 404 422

query_id 500

PUT /saved-queries/{query_id}

query_id

title

natural_language_query

generated_code

result_summary

```
curl -X PUT http://localhost:8000/saved-queries/1 \
  -H "X-API-Key: insightagent-secret-key" \
  -H "Content-Type: application/json" \
  -d '{"title": "Updated Title"}'
```

200

404

```
{
  "detail": "SavedQuery 999 not found."
}
```

401 404 422 500

DELETE /saved-queries/{query_id}

query_id

```
curl -X DELETE http://localhost:8000/saved-queries/1 \
  -H "X-API-Key: insightagent-secret-key"
```

204

404

```
{
  "detail": "SavedQuery 999 not found."
}
```

401 404 500

X-API-Key

GET /analytics/store-sales-summary

```
curl http://localhost:8000/analytics/store-sales-summary \
  -H "X-API-Key: insightagent-secret-key"
```

200

```
[
  {
    "city": "San Francisco",
    "country": "USA",
    "total_revenue": 1234567.89,
    "order_count": 150
  }
]
```

401 500

GET /analytics/product-ranking

limit	20	1 – 100

```
curl "http://localhost:8000/analytics/product-ranking?limit=5" \
  -H "X-API-Key: insightagent-secret-key"
```

200

```
[
  {
    "productCode": "S18_3232",
    "productName": "1992 Ferrari 360 Spider red",
    "productLine": "Classic Cars",
    "total_revenue": 276839.98,
    "total_quantity": 1808
  }
]
```

401 422 limit 500

GET /analytics/employee-performance

```
curl http://localhost:8000/analytics/employee-performance \
  -H "X-API-Key: insightagent-secret-key"
```

200

```
[
  {
    "employeeNumber": 1370,
    "firstName": "Gerard",
    "lastName": "Hernandez",
    "jobTitle": "Sales Rep",
    "total_revenue": 1258577.81,
    "order_count": 14,
    "customer_count": 7
```

```
  }
]
```

401 500

GET /analytics/sales-trend

---

year	null	2004

```
# All years
curl http://localhost:8000/analytics/sales-trend \
  -H "X-API-Key: insightagent-secret-key"

# Year 2004 only
curl "http://localhost:8000/analytics/sales-trend?year=2004" \
  -H "X-API-Key: insightagent-secret-key"
```

200

```
[
  {
    "year": 2004,
    "month": 11,
    "monthly_revenue": 1029837.66,
    "order_count": 29
  }
]
```

401 422 year 500

POST /analytics/analyze

observe_schema

run_python_analysis

final_answer

saved_queries

---

question	✅

```
curl -X POST http://localhost:8000/analytics/analyze \
  -H "X-API-Key: insightagent-secret-key" \
  -H "Content-Type: application/json" \
  -d '{"question": "Which product line generated the most revenue?"}'
```

200

```
{
  "answer": "Classic Cars generated the highest revenue at
        $3,853,438.94, followed by Vintage Cars at $1,903,150.84.",
  "iterations": 3,
  "generated_code": "import pandas as pd\ndf = read_sql('SELECT
        productLine, SUM(quantityOrdered * priceEach) AS revenue
        ...')\nprint(df.to_csv(index=False))",
  "saved_query_id": 12,
  "log_id": 5,
  "tool_trace": ["observe_schema", "run_python_analysis",
        "final_answer"]
}
```

---

answer	
iterations	
generated_code	
saved_query_id	saved_queries
log_id	analysis_logs

tool_trace	["observe_schema", "run_python_analysis", "final_answer"]

question	500	401 422

GET /analytics/logs

analysis_logs

skip	0	≥ 0
limit	20	1 – 100

```
curl http://localhost:8000/analytics/logs \
  -H "X-API-Key: insightagent-secret-key"
```

200

```
[
  {
    "id": 5,
    "query_text": "Which product line generated the most revenue?",
    "status": "completed",
    "iterations": 3,
    "final_answer": "Classic Cars generated the highest revenue...",
    "created_at": "2025-04-13 10:00:00"
  }
]
```

401 422 500

GET /analytics/logs/{log_id}/turns

log_id	POST /analytics/analyze → log_id

```
curl http://localhost:8000/analytics/logs/5/turns \
  -H "X-API-Key: insightagent-secret-key"
```

200


```
[
  {
    "id": 1,
    "iteration": 1,
    "tool_name": "observe_schema",
    "llm_content": "I need to understand the database structure
        first.",
    "tool_input": "{}",
    "tool_output": "{\"database\": \"enterprise_api\", \"tables\":
        [...]}",
    "created_at": "2025-04-13 10:00:01"
  },
  {
    "id": 2,
    "iteration": 2,
    "tool_name": "run_python_analysis",
    "llm_content": null,
    "tool_input": "{\"code\": \"import pandas as pd\\ndf =
        read_sql(...)\"}",
    "tool_output": "{\"status\": \"success\", \"stdout\":
        \"productLine,revenue\\nClassic Cars,3853438.94\"}",
    "created_at": "2025-04-13 10:00:04"
  },
  {
    "id": 3,
    "iteration": 3,
    "tool_name": "final_answer",
    "llm_content": "Classic Cars generated the highest revenue at
        $3,853,438.94.",
    "tool_input": null,
    "tool_output": null,
    "created_at": "2025-04-13 10:00:06"
  }
]
```


404

```
{
  "detail": "Analysis log 999 not found."
}
```

401 404 422
log_id 500

X-API-Key

POST /skills/observe_schema

enterprise_api
app_ro

```
curl -X POST http://localhost:8000/skills/observe_schema \
  -H "X-API-Key: insightagent-secret-key"
```

200

```
{
  "database": "enterprise_api",
  "tables": [
    {
      "table_name": "employees",
      "columns": [
        {
          "name": "employeeNumber",
          "declared_type": "INT",
          "nullable": false,
          "is_primary_key": true
        },
        {
          "name": "lastName",
          "declared_type": "VARCHAR(50)",
          "nullable": false,
          "is_primary_key": false
        }
```

```
      ],
      "foreign_keys": [
        {
          "from_column": "officeCode",
          "ref_table": "offices",
          "ref_column": "officeCode"
        },
        {
          "from_column": "reportsTo",
          "ref_table": "employees",
          "ref_column": "employeeNumber"
        }
      ]
    }
  ]
}
```


500


```
{
  "detail": {
    "status": "failed",
    "error_type": "SCHEMA_READ_ERROR",
    "message": "Failed to inspect database schema."
  }
}
```


401 500

POST /skills/run_python_analysis

- read_sql(sql, params=(), max_rows=50000) pandas.DataFrame
- pandas pd numpy np math statistics datetime re
- print()
-

```
Input code
    │
    ▼
AST Guard  ──(blocked)──► SECURITY_VIOLATION error
    │ pass
    ▼
SQL Guard  ──(blocked)──► SQL_BLOCKED error
```

```
    │ pass
    ▼
Subprocess Worker (isolated process, app_ro DB account)
    │
    ▼
Structured JSON result
```

os sys subprocess socket requests pathlib shutil open() eval() exec() compile() __import__()

SELECT WITH INSERT UPDATE DELETE DROP

code	✅

```
curl -X POST http://localhost:8000/skills/run_python_analysis \
  -H "X-API-Key: insightagent-secret-key" \
  -H "Content-Type: application/json" \
  -d '{
    "code": "import pandas as pd\ndf = read_sql(\"SELECT productLine,
        SUM(quantityOrdered * priceEach) AS revenue FROM orderdetails
        od JOIN products p ON od.productCode = p.productCode GROUP BY
        productLine ORDER BY revenue
        DESC\")\nprint(df.to_csv(index=False))"
  }'
```

200

```
{
  "status": "success",
  "stdout": "productLine,revenue\nClassic
        Cars,3853438.9400000002\nVintage Cars,1903150.84\n...",
  "data_found": true,
  "summary": {
    "lines": 7,
    "chars": 210
  }
}
```

> data_found true stdout false

403

```
{
  "detail": {
    "status": "blocked",
    "error_type": "SECURITY_VIOLATION",
    "message": "Import of module 'os' is not allowed.",
    "hint": "Use only pandas, numpy, math, statistics, datetime, re."
  }
}
```

403

```
{
  "detail": {
    "status": "failed",
    "error_type": "SQL_BLOCKED",
    "message": "Only SELECT / WITH statements are allowed."
  }
}
```

422

```
{
  "detail": {
    "status": "blocked",
    "error_type": "SYNTAX_ERROR",
    "message": "SyntaxError: invalid syntax (line 3)"
  }
}
```

422

```
{
  "detail": {
    "status": "failed",
    "error_type": "RUNTIME_ERROR",
    "message": "ZeroDivisionError: division by zero"
  }
}
```

408

```
{
  "detail": {
    "status": "failed",
    "error_type": "EXECUTION_TIMEOUT",
    "message": "Execution exceeded 30 seconds.",
    "hint": "Simplify the query or reduce the dataset size."
  }
}
```

500

```
{
  "detail": {
    "status": "failed",
    "error_type": "WORKER_ERROR",
    "message": "Internal subprocess error."
  }
}
```

401 403 408
422 code 500

---

tool_trace POST /analytics/analyze

---

observe_schema

run_python_analysis

final_answer

---

observe_schema → run_python_analysis → run_python_analysis → final_answer

final_answer

GET /analytics/logs/{log_id}/turns